sPHENIX Self-Paced Course — Module 3

Calorimetry in sPHENIX: EMCal and HCal hardware

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Self-contained study text built on the Summer-2026 sPHENIX School, public sPHENIX papers and standard calorimetry textbooks

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How to use this module

This module is self-contained. Everything you need is explained in the text, from first principles to the way sPHENIX actually does it. You do not need to open slides, recordings or papers to follow it. The school talks listed below are the *sources* the module was built from; they are cited as [S##] so that you can trace a statement if you ever want to.

Learning objectives

After this module you should be able to:

O1. Explain how electromagnetic and hadronic showers develop, using the radiation length X_0, the Molière radius R_M, the critical energy E_c and the nuclear interaction length λ_I.

O2. Explain what a sampling calorimeter is, define the sampling fraction, and estimate the stochastic term from the sampling geometry.

O3. Describe the sPHENIX EMCal (tungsten-powder/scintillating-fibre SPACAL blocks, 2D projectivity, light guides, SiPMs) and its tower grid of 96×256 in $\eta \times \phi$.

O4. Describe the inner and outer HCal (aluminium frame and steel flux return, tilted plates, scintillating tiles with wavelength-shifting fibres, 24×64 towers per layer).

O5. Follow the signal chain from photon in the fibre to a calibrated `TowerInfo` object: SiPM, preamplifier, digitiser, waveform, template fit, status, calibration.

O6. Decompose the energy resolution into stochastic, constant and noise terms, and say which hardware or calibration effect feeds each term.

O7. Outline the calibration chain: test beam, LED and cosmic tests, cosmic-muon MIP scale for the HCal, tower-slope relative calibration, π^0 calibration for the EMCal, temperature and timing corrections.

O8. Connect all of the above to the π^0 and η peaks you measure in your own analysis.

Prerequisites and time

Modules 1 (physics programme) and 2 (RHIC and detector overview). Estimated time: 9–11 hours (about 6 hours of reading, 2 hours for the worked examples, 1–3 hours for the optional exercises).

Sources (sPHENIX School 2026, BNL, July 2026)

- [S09] C. Woody, “EMCal Overview” — the main EMCal hardware source.
- [S08] J. Haggerty, “HCal and Global Detector Overview” — HCal design, tiles, test beams, LEDs, lessons learned.
- [S47] B. Seidlitz, J. Nagle, S. Bathe, “Calorimeter test-beam simulation” (workfest) — how to read test-beam results.
- [S42] A. M. Khan, “HCal Calibration studies” (flash talk) — the tower-slope method.
- Also used for the calibration overview: [S15] B. Seidlitz, “EMCal Reco and Calibrations”; [S16] A. Menon, “HCal Reco and Calibrations”. Module 7 treats reconstruction in depth.

Public readings (optional)

Aidala et al., IEEE TNS 65 (2018) 2901, arXiv:1704.01461 (prototype design and beam tests) [R1]; Aidala et al., IEEE TNS 68 (2021) 173, arXiv:2003.13685 (2D-projective EMCal prototype) [R2]; the sPHENIX proposal, arXiv:1501.06197 [R3]; the PDG review of particle detectors and passage of particles through matter [R4].

Notation used throughout

Energies in GeV unless stated. $\oplus$ means addition in quadrature: $a \oplus b = \sqrt{a^2 + b^2}$. "Tower" means the smallest read-out cell of a calorimeter. "Block" (EMCal) means one machined tungsten/fibre piece containing 2×2 towers. η is pseudorapidity, ϕ azimuth; "$i\eta$, $i\phi$" are integer tower indices.

1 Why calorimetry, and what sPHENIX needed

Goal. Understand which physics measurements set the calorimeter requirements, and why they lead to a compact, projective, finely segmented EMCal and a two-layer HCal built into the magnet return yoke.

1.1 What a calorimeter does

A calorimeter measures the energy of a particle by stopping it completely. The particle interacts with dense material and produces a cascade (a "shower") of lower-energy secondaries. Eventually all of the energy is converted into ionisation and excitation of atoms, plus some invisible losses. A fraction of that deposited energy is turned into a measurable signal, here scintillation light. The signal is proportional to the incident energy, so one number (the calibration constant) converts signal to energy.

Calorimeters have three properties that trackers do not have. First, they measure neutral particles (photons, neutrons, K^0_L). Second, their relative resolution *improves* with energy, as $1/\sqrt{E}$, while a tracker's momentum resolution degrades as p grows. Third, they are fast and can be used to trigger: a large energy deposit is a simple signature for a high-p_T photon or jet (Module 5).

There are two kinds of showers, electromagnetic (photons, electrons) and hadronic (pions, protons, neutrons, ...). There are two kinds of calorimeters: *homogeneous* (the whole volume is active, e.g. a lead-glass or crystal block) and *sampling* (layers or fibres of active material embedded in a passive, dense absorber). Both sPHENIX calorimeters are sampling calorimeters [S09, S08].

1.2 Physics drivers

The sPHENIX calorimeter system was designed around three classes of measurements [S09, S08, R3]:

1. **Jets** in p+p, p+Au and Au+Au. Jets need both the electromagnetic part (mostly photons from π^0 decays) and the hadronic part of the energy. This drives full azimuthal coverage, $|\eta| < 1.1$, and a hadronic calorimeter.
2. **Direct photons and** π^0 (your physics). These need an EMCal with good energy resolution, a small shower size and fine segmentation.
3. $\Upsilon \to e^+e^-$. Electrons of about 5 GeV must be identified against a large hadron background, using E/p (EMCal energy over track momentum) and shower shape.

The energy range is modest compared with the LHC: roughly 1 to 100 GeV for single particles and roughly 20 to 70 GeV for jets [S08]. The design goals were an EMCal resolution of about $15\%/\sqrt{E}$ and a single-hadron HCal resolution of order $100\%/\sqrt{E}$ [S09, R3].

1.3 The underlying event sets the scale

In central Au+Au collisions thousands of soft particles are produced. Every tower receives some energy from this "underlying event" (UE), and that energy fluctuates. For a photon or electron cluster, what matters is the UE energy inside the cluster area and its fluctuation. The smaller the cluster area, the smaller this background. Hence the EMCal must have a small Molière radius and small towers.

Worked example: UE energy under a cluster (toy)

Take a transverse-energy density at midrapidity of order $\mathrm{d}E_T/\mathrm{d}\eta \approx 600$ GeV in the most central Au+Au events (the order of magnitude measured at RHIC; see the public sPHENIX E_T paper [R5] for the actual values). Spread uniformly in ϕ, this is $600/(2\pi) \approx 95$ GeV per unit of η per radian. A 3×3 EMCal tower array covers $\Delta\eta \times \Delta\phi = 0.075 \times 0.074 \approx 0.0055$. The UE then deposits roughly $95 \times 0.0055 \approx 0.5$ GeV in the cluster area (only part of it in the EMCal, since hadrons pass through). Compare: a 5 GeV electron measured with $15\%/\sqrt{E}$ has $\sigma_E \approx 0.15\sqrt{5} \approx 0.34$ GeV. *Conclusion:* the UE contribution is comparable to the detector resolution. Making the cluster area four times larger would make the UE dominate. This is why the EMCal is compact and finely segmented. (Toy numbers; illustrative only.)

1.4 Mechanical constraints

sPHENIX re-uses the former BaBar superconducting solenoid (1.4 T) [S09, R3]. Early designs put the solenoid inside the calorimeters; the final design places the EMCal and the inner HCal *inside* the solenoid and the outer HCal outside it [S09]. Three consequences follow:

- The EMCal sits close behind the tracker, with little material in front. Photons have less chance to convert or shower before the EMCal.
- The EMCal inner radius is about 90 cm, set by the need for space for the tracker and by occupancy in Au+Au; the outer radius is about 116 cm. The EMCal therefore has only about 26 cm of radial space, including electronics, so it must be very dense [S09].
- The outer HCal steel doubles as the magnet's flux return. This is economical: the return yoke is needed anyway, so instrumenting it gives a hadronic calorimeter at low cost [S08, S09].

1.5 Projectivity

A tower is *projective* if its long axis points back to the interaction point. A particle from the vertex then travels along the tower axis, and its shower stays in few towers. In a non-projective geometry, a particle at large $|\eta|$ enters at an angle; its shower crosses several towers in depth and looks wider. sPHENIX chose an approximately *two-dimensional* projective EMCal: towers are tilted in both η and ϕ [S09]. At $\eta \approx 0$, a one-dimensional projective design would be almost as good. At $|\eta| \to 1$ the 2D projectivity keeps the transverse shower size small, which improves energy resolution and electron–hadron separation in the high-multiplicity environment [S09, R2].

Checks

You should be able to answer: Why is the EMCal inside the magnet? (Less material in front; compact.) Why must it be dense? (Little radial space, small showers for UE.) Why does the HCal live in the return yoke? (Cost and space.)

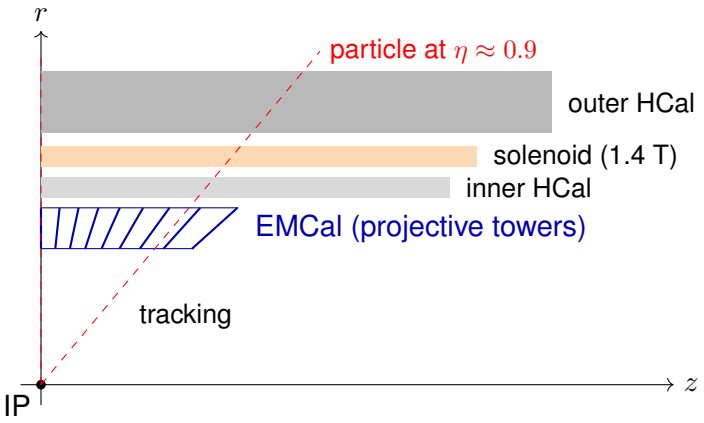


Figure 1: Schematic r–z quarter view (not to scale). EMCal towers point back towards the interaction point (IP). The inner HCal and the EMCal are inside the solenoid; the outer HCal is outside and acts as the flux return.

2 Electromagnetic showers

Goal. Be able to estimate, with pencil and paper, where a photon shower starts, where it peaks, how deep and how wide it is, and how much of it a block of given size contains.

2.1 Interactions of electrons and photons

Above a few tens of MeV, electrons and positrons lose energy mainly by *bremsstrahlung*: they radiate photons in the Coulomb field of nuclei. Photons above a few MeV interact mainly by *pair production*, $\gamma \to e^+e^-$, again in the nuclear field. Both processes are governed by the same material constant, the **radiation length** X_0.

- An electron loses on average a fraction $1 - 1/e$ of its energy by bremsstrahlung in one X_0: $\langle E(x) \rangle = E_0\, e^{-x/X_0}$.
- A high-energy photon has a mean free path for pair production of $\frac{9}{7}X_0$.

A useful approximation for X_0 is [R4]

$$X_0 \approx \frac{716.4\ \mathrm{g\,cm^{-2}}\ A}{Z(Z+1)\ln(287/\sqrt{Z})}. \tag{1}$$

It falls roughly as $1/Z^2$, so high-Z materials such as tungsten ($Z = 74$) or lead ($Z = 82$) are compact. For tungsten this gives about 6.8 g/cm^2, i.e. 0.35 cm at a density of 19.3 g/cm^3.

Below the **critical energy** E_c, ionisation losses exceed radiation losses, and the electrons simply range out. Roughly $E_c \approx 610\ \mathrm{MeV}/(Z + 1.24)$ for solids; for tungsten $E_c \approx 8$ MeV [R4].

2.2 The Heitler toy model

A simple model captures the essentials. Assume that after every X_0 each particle splits into two of half the energy (an electron radiates one photon; a photon converts into a pair). After t radiation lengths there are $N(t) = 2^t$ particles of energy $E_0/2^t$. Multiplication stops when the energy per particle reaches E_c. This gives

$$t_{\max} = \frac{\ln(E_0/E_c)}{\ln 2}, \qquad N_{\max} = \frac{E_0}{E_c}. \tag{2}$$

Two lessons follow. The depth of the shower maximum grows only *logarithmically* with energy, so a calorimeter of fixed depth contains showers over a wide energy range. The number of charged particles, and hence the signal, is *proportional* to E_0, which is the basis of calorimetry. The total track length of charged particles is $\propto E_0/E_c \cdot X_0$; fluctuations in this number set the ultimate (intrinsic) resolution.

2.3 Longitudinal profile

A better description, fitted to simulations, is a gamma distribution [R4]:

$$\frac{1}{E_0}\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{b\,(bt)^{a-1}e^{-bt}}{\Gamma(a)}, \qquad t = x/X_0, \qquad t_{\max} = \frac{a-1}{b}, \tag{3}$$

with $b \approx 0.5$ and

$$t_{\max} \approx \ln\frac{E_0}{E_c} + C_j, \qquad C_e = -0.5, \quad C_\gamma = +0.5. \tag{4}$$

Photons shower about one X_0 later than electrons, because the photon must first convert. The fraction of energy contained in a depth T is the regularised incomplete gamma function $P(a, bT)$.

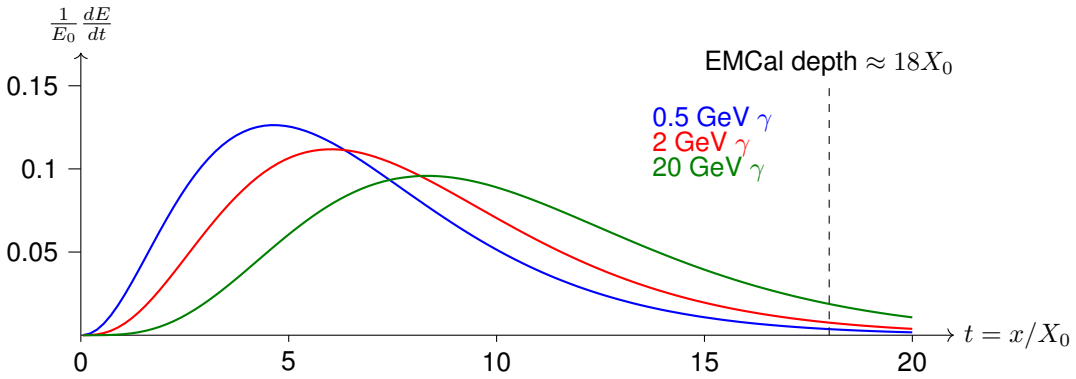


Figure 2: Toy longitudinal profiles from the gamma-function parametrisation with $b = 0.5$ and $E_c = 8$ MeV (pure-tungsten value). Schematic; not sPHENIX data.

2.4 Lateral profile and the Molière radius

The shower spreads sideways mainly through multiple Coulomb scattering of the low-energy electrons. The natural transverse scale is the **Molière radius**

$$R_M \approx X_0\,\frac{21.2\ \mathrm{MeV}}{E_c}. \tag{5}$$

About 90% of the energy lies inside a cylinder of radius $1\,R_M$ around the shower axis, and about 95% inside $2\,R_M$ [R4]. The lateral profile has a narrow core (from the early, energetic part of the shower) and a broad halo (from low-energy photons travelling far before converting). R_M is nearly independent of the incident energy, and it depends on $X_0/E_c \propto A/Z$ rather than on Z^2, so it is less sensitive to the material than X_0.

2.5 Numbers for the sPHENIX EMCal

The sPHENIX EMCal absorber is a mixture of tungsten powder and epoxy with embedded scintillating fibres. Its density is about 9–10 g/cm^3, giving $X_0 \approx 7$ mm and $R_M \approx 2.3$ cm, with a total depth of about 18 X_0 [S09, R1]. The mixture has a lower average Z than pure tungsten, so its E_c is somewhat higher

than 8 MeV and showers peak slightly earlier than the pure-tungsten estimate below. The estimate is therefore conservative.

Worked example: containment of a 2 GeV photon (checkpoint exercise)

Longitudinal. $t_{\max} = \ln(2000/8) + 0.5 = 5.5 + 0.5 = 6.0\,X_0$, i.e. about 4.2 cm into the block. With $b = 0.5$, $a = 1 + b\,t_{\max} = 4.0$. The fraction inside $T = 18\,X_0$ is $P(4.0,\,9) = 1 - e^{-9}(1+9+9^2/2+9^3/6) \approx 0.98$. So about 2% leaks out of the back. For 20 GeV ($a \approx 5.2$) the leakage grows to about 6%; for 0.5 GeV it is about 1%.

Lateral. The tower front face is about $\Delta\phi \times R = 2\pi/256 \times 93\ \text{cm} \approx 2.3$ cm, i.e. about $1\,R_M$. A 3×3 array centred on the shower extends ± 1.5 towers $\approx \pm 1.5\,R_M$ from the axis. Interpolating between the 90% ($1R_M$) and 95% ($2R_M$) rules gives about 92–93% containment (somewhat less if the photon hits a tower corner, since the 3×3 is then off-centre).

Total. Roughly $0.98 \times 0.93 \approx 0.91$ of the energy is inside a 3×3 array. A 5×5 array ($\pm 2.5 R_M$) recovers about 96–97% laterally.

Lesson. The "raw" cluster energy is several percent below the true energy, and the missing fraction depends on energy and impact position. The energy calibration (Section 10) and the simulation must account for it consistently. (Approximate textbook model; not a sPHENIX measurement.)

2.6 Two photons from a π^0

For a π^0 of energy E, the minimum opening angle of the two decay photons is $\theta_{\min} \approx 2m_{\pi^0}/E$. At $E = 5$ GeV this is 0.054 rad, i.e. about 5 cm at $R = 93$ cm, or about two tower widths. At $E = 10$ GeV it is about one tower width. At higher energy the two showers merge into one cluster. The small R_M and the small towers of the sPHENIX EMCal push this merging threshold up; above it, π^0 identification relies on shower shape rather than on two separate clusters.

Connect to your analysis

Your $\pi^0/\eta \to \gamma\gamma$ reconstruction uses the invariant mass $m^2 = 2E_1E_2(1 - \cos\theta_{12})$. Lateral leakage outside the cluster lowers E_1 and E_2, and so shifts the reconstructed peak below m_{π^0} unless the calibration absorbs it. Because the merging of the two photons sets in at high p_T, your π^0 efficiency from PYTHIA8 with EMCal smearing must reproduce the cluster splitting/merging; this is one reason the simulation must model the shower width correctly. The η meson ($m_\eta \approx 548$ MeV, four times heavier) opens up four times more, so it merges at much higher energy.

Pitfalls

- Using the pure-tungsten X_0 (3.5 mm) for the EMCal: the powder/epoxy/fibre mixture has $X_0 \approx 7$ mm.
- Forgetting that photons shower about one X_0 later than electrons.
- Treating lateral containment as energy-independent: the core is nearly energy-independent, but the leakage of the halo and of the back of the shower is not.

3 Sampling calorimetry and energy resolution

Goal. Know what sets the energy resolution of a sampling calorimeter, be able to estimate each term, and know which term each hardware or calibration choice affects.

3.1 Sampling fraction

In a sampling calorimeter only the active medium produces a signal. The **sampling fraction** is

$$f_{\mathrm{samp}} = \frac{E_{\mathrm{dep}}(\text{active})}{E_{\mathrm{dep}}(\text{active}) + E_{\mathrm{dep}}(\text{absorber})}. \tag{6}$$

For a minimum-ionising particle (MIP) it can be computed from the $\mathrm{d}E/\mathrm{d}x$ and thicknesses of the two media. For electromagnetic showers the actual value is usually somewhat smaller than the MIP value, because the many soft photons in a shower are absorbed preferentially in the high-Z absorber (photoelectric effect scales as Z^{4-5}). The sPHENIX EMCal has $f_{\mathrm{samp}} \approx 2\%$: only about one fiftieth of the shower energy is deposited in the fibres [S09, R1]. The HCal sampling fraction is a few percent and, because the scintillating tiles are flat while the steel plates are tapered, it changes with depth [S08].

3.2 The resolution formula

The relative energy resolution of a calorimeter is written as

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b \oplus \frac{c}{E}. \tag{7}$$

- **Stochastic term a.** Any Poisson-like fluctuation in the number of independent signal quanta $N \propto E$ gives $\sigma_N/N = 1/\sqrt{N} \propto 1/\sqrt{E}$. In a sampling calorimeter the dominant sources are *sampling fluctuations* (how many shower particles happen to cross the active medium) and *photostatistics* (how many photoelectrons are produced).
- **Constant term b.** Effects proportional to E: tower-to-tower calibration errors, non-uniform response across the tower face or along the depth, leakage that fluctuates proportionally. This term dominates at high energy.
- **Noise term c.** An energy-independent noise σ_{noise} (electronics, pedestal width, pile-up, underlying event) gives $\sigma_E/E = \sigma_{\mathrm{noise}}/E$. It dominates at low energy.

The terms add in quadrature because they are independent.

3.3 Estimating the sampling term

A widely used empirical rule for electromagnetic sampling calorimeters [R6] is

$$\frac{a_{\mathrm{samp}}}{\sqrt{E}} \approx 2.7\% \sqrt{\frac{d\,[\mathrm{mm}]}{f_{\mathrm{samp}}}}\; \frac{1}{\sqrt{E\,[\mathrm{GeV}]}}, \tag{8}$$

where d is the thickness of the active layers (here, the fibre diameter). The reasoning: the number of independent crossings of the active medium is proportional to $E\, f_{\mathrm{samp}}/d$; finer sampling (smaller d) at the same f_{samp} gives more crossings and smaller fluctuations.

Worked example: sampling term of the sPHENIX EMCal

Fibres of 0.47 mm diameter and $f_{\mathrm{samp}} \approx 0.02$ give $a_{\mathrm{samp}} \approx 2.7\%\sqrt{0.47/0.02} = 2.7\% \times 4.85 \approx 13\%$. Add photostatistics: if a tower produces N_{pe} photoelectrons per GeV, the contribution is $1/\sqrt{N_{\mathrm{pe}}}$. For a toy value of 400 p.e./GeV this is 5%. Then $a \approx 13\% \oplus 5\% \approx 14\%$. This is close to the 15% design goal, which shows that the fibre geometry, not the photodetector, sets the stochastic term. (Rule-of-thumb estimate.)

3.4 Measured resolutions in beam tests

The prototypes were measured at the Fermilab Test Beam Facility (experiment T-1044) between 2014 and 2018 [S09, S08, R1, R2]. Public results:

- 2016 prototype at $\eta \approx 0$, electrons in the EMCal: $\sigma_E/E = 2.8\% \oplus 15.5\%/\sqrt{E}$ [R1].
- Same campaign, hadrons in the combined EMCal+HCal: $\sigma_E/E = 13.5\% \oplus 64.9\%/\sqrt{E}$ [R1].
- 2018 2D-projective EMCal prototype at $\eta \approx 1$: $3.5\% \oplus 13.3\%/\sqrt{E}$ or $3.0\% \oplus 15.4\%/\sqrt{E}$, depending on the position-dependent correction used [R2].

Worked example: what $2.8\% \oplus 15.5\%/\sqrt{E}$ means for your photons

E (GeV)	1	2	5	10	20	30
σ_E/E (%)	15.8	11.3	7.5	5.6	4.5	4.0
σ_E (GeV)	0.16	0.23	0.37	0.56	0.89	1.19

At 2 GeV the stochastic term is $15.5/\sqrt{2} = 11.0\%$ and dominates; at 30 GeV it is 2.8%, equal to the constant term. For most π^0 photons in p+p (a few GeV) the stochastic term matters most; at the highest p_T the constant term, i.e. calibration quality, takes over.

3.5 From photon resolution to π^0 mass resolution

Differentiate $m^2 = 2E_1E_2(1-\cos\theta)$:

$$\frac{\sigma_m}{m} = \frac{1}{2}\left[\frac{\sigma_{E_1}}{E_1} \oplus \frac{\sigma_{E_2}}{E_2} \oplus \frac{\sigma_\theta}{\tan(\theta/2)}\right]. \tag{9}$$

For a symmetric decay with $E_1 = E_2 = 2$ GeV, the energy part alone gives $\frac{1}{2}\sqrt{2} \times 11.3\% \approx 8\%$ of m_{π^0}, about 11 MeV. The angular term adds to this; it comes from the cluster position resolution, which in turn depends on tower size and shower shape. In real data the peak is also broadened by the underlying event (Au+Au), by miscalibration (constant term), and by non-linearity.

Connect to your analysis

The width of your π^0 peak is a direct read-out of the photon resolution. In your PYTHIA8 efficiency, the "EMCal smearing" is the extra Gaussian smearing added to simulated cluster energies (and positions) until the simulated π^0 and η widths match data [S15]. If the smearing is too small, the simulated efficiency for a given mass window is too high. If your analysis uses a fixed mass window, a mismatch in width between data and MC is a direct systematic on the cross section.

3.6 Linearity

A calorimeter is linear if the mean signal is proportional to the incident energy. Sources of non-linearity in sPHENIX are (i) SiPM saturation, since each SiPM has a finite number of pixels (Section 5); (ii) the readout electronics range; (iii) energy-dependent leakage [S09]. The EMCal prototypes showed a measurable non-linearity at high energy, attributed to SiPM saturation and electronics [S09, R2]. The non-linearity affects exactly the high-energy region where single photons are measured, so it must be modelled.

Pitfalls

- Reading a test-beam "resolution vs energy" plot without checking how each point was normalised. If each energy point was rescaled so that its mean equals the beam energy, the plot only tests the *shape* (width) of the response, and tells you nothing about linearity or the absolute data-to-MC scale [S47]. Linearity must be shown with a single energy-independent scale factor.
- Comparing resolutions with different beam spot sizes: a small beam spot centred on a tower gives a better resolution than a beam spread over the tower face, because of the position dependence of the response [S09, R2].
- Forgetting that $\oplus$ is quadrature: $2.8\% \oplus 15.5\%/\sqrt{E}$ is *not* $2.8\% + 15.5\%/\sqrt{E}$.

4 The sPHENIX EMCal

Goal. Know how the EMCal is built, from fibre to sector, and why each choice was made.

4.1 The SPACAL concept

A *spaghetti calorimeter* (SPACAL) is a sampling calorimeter in which the active medium is a dense array of scintillating fibres running (nearly) along the particle direction, embedded in an absorber [S09]. The idea was developed in the 1980s (e.g. by the SPACAL collaboration, with lead absorber) and used in several experiments. Earlier SPACALs grooved lead plates and laid fibres into them. sPHENIX introduced a new variant: a matrix of **tungsten powder and epoxy**, cast around the fibres [S09]. This is denser than lead SPACAL, so the calorimeter fits into the available 26 cm of radius. The sPHENIX EMCal was the first W/SciFi SPACAL deployed in a running experiment [S09].

4.2 Blocks, towers and sectors

- **Fibres:** 0.47 mm diameter, about 1 mm apart, from two vendors [S09].
- **Block fabrication:** a mould holds the fibre array (positioned by metal meshes along its length); tungsten powder is poured in and the mould is infused with epoxy; the block is then machined to its final tapered shape [S09].
- **Block = 2×2 towers.** Each block is read out as four towers [S09, R2].
- **Totals:** 6144 blocks $\times$ 4 = 24 576 towers, arranged as 96 towers in η times 256 towers in ϕ, covering $|\eta| < 1.1$ and 2π [S09].
- **Tower size:** $\Delta\eta \times \Delta\phi \approx 0.025 \times 0.025$ ($2\pi/256 = 0.0245$ rad).
- **Sectors:** 64 sectors, two in η (north/south half) times 32 in ϕ [S09]. Each sector covers 8 towers in ϕ ($256/32$) and 48 in η ($96/2$).
- **Production:** the mid-rapidity blocks ($|\eta| < 0.85$) were produced at UIUC, and the high-rapidity blocks ($0.85 < |\eta| < 1.1$) in China. Modules and sectors were assembled and tested at BNL; installation ran from 2022 [S09].

4.3 Light collection

Scintillation light travels along each fibre to the readout end, which faces the interaction point (front readout) [S09]. At the end of each tower sits a short acrylic *light guide* that funnels light onto four 3×3 mm^2 SiPMs, whose signals are summed passively [S09]. Two consequences matter for analysis:

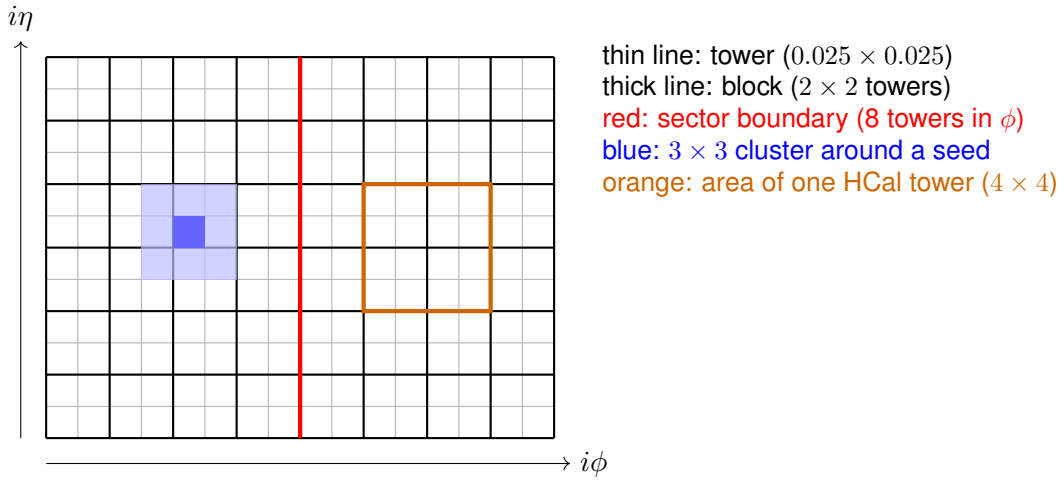


Figure 3: Schematic patch of the EMCal tower grid (checkpoint exercise). The full grid is $96\ (i\eta) \times 256\ (i\phi)$; each HCal tower ($0.1 \times 0.1$) covers 4×4 EMCal towers. Not to scale; the actual towers are tilted for projectivity.

- The SiPMs cover only a small fraction of the light-guide exit area, and a short light guide mixes the light poorly. The response therefore depends somewhat on *where* in the tower face the shower hits [S09].
- Total light collection is still fairly uniform, but the residual *position dependence* must be corrected (in beam tests with a position-dependent correction, in sPHENIX through the clustering and calibration) [S09, R2].

The resulting photostatistics contribute a few percent divided by $\sqrt{E}$ to the resolution, well below the sampling term [S09]. For future detectors (the ePIC forward EMCal) larger SiPMs covering more of the readout face improve the light collection; the sPHENIX experience led directly to that design [S09].

4.4 Testing before installation

After a burn-in period, each sector was tested at BNL with LED pulses (to check every channel and its gain) and with cosmic rays (to see the MIP peak in each tower), with the sector rotated in a test stand [S09]. These tests gave the first relative calibration and the list of bad channels before installation.

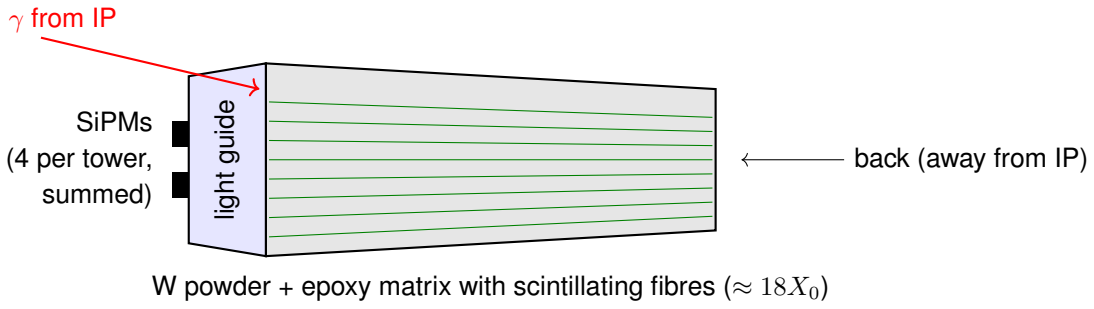


Figure 4: Schematic side view of one EMCal tower: fibres run along the tower, the light exits at the front (towards the IP) through a short light guide onto SiPMs. Not to scale.

Checks

You should be able to reproduce: $24\,576 = 96 \times 256$; 6144 blocks; 0.025×0.025 tower size; 64 sectors; one HCal tower above 4×4 EMCal towers. If you see 128 or 192 in $i\phi$ in a plot, you are probably looking at the wrong detector or a merged-tower view.

Pitfalls

- The towers are tilted (projective). A shower from a vertex far from $z = 0$ enters the towers at an angle, so the shower is wider and the depth of shower maximum maps to a different η. Vertex-dependent corrections exist for this reason (Module 7).
- Sector and block boundaries have small dead regions and different response; acceptance maps in $(i\eta, i\phi)$ show this structure.

5 Silicon photomultipliers

Goal. Understand the SiPM well enough to explain saturation, temperature dependence, dark noise and radiation damage, all of which appear in sPHENIX calibration and QA.

5.1 How a SiPM works

A SiPM is an array of thousands of tiny avalanche photodiodes (pixels or “microcells”), all connected in parallel [S09]. Each pixel is reverse-biased *above* its breakdown voltage V_{bd} (Geiger mode). One photoelectron in a pixel triggers a self-sustaining avalanche, which is stopped (quenched) by a series resistor. Each fired pixel delivers the same charge $Q = C_{\mathrm{pix}}(V_{\mathrm{bias}} - V_{\mathrm{bd}})$. The SiPM output is the sum of the fired pixels, so it counts photons. Key properties:

- Gain of order 10^5–10^6, so remote digitisation of the signal is possible [S09, S08].
- Photon detection efficiency (PDE) of about 25% at the scintillator wavelength [S09].
- Insensitive to magnetic fields, unlike photomultiplier tubes. This is essential inside a 1.4 T solenoid.
- Small, cheap and low-power [S08].

The EMCal uses Hamamatsu MPPCs with 15 μm pixels on a 3×3 mm^2 area [S09, R1], i.e. $(3000/15)^2 = 40\,000$ pixels each.

5.2 Saturation

If N_{pe} photoelectrons arrive at a SiPM with N_{pix} pixels spread uniformly and at once, the expected number of fired pixels is

$$N_{\mathrm{fired}} = N_{\mathrm{pix}} \left[1 - \exp\!\left(-\frac{N_{\mathrm{pe}}}{N_{\mathrm{pix}}} \right) \right] \approx N_{\mathrm{pe}} \left(1 - \frac{N_{\mathrm{pe}}}{2 N_{\mathrm{pix}}} \right). \tag{10}$$

The response is linear only for $N_{\mathrm{pe}} \ll N_{\mathrm{pix}}$. If the light illuminates only part of the SiPM (poor light mixing), the effective number of pixels is smaller and saturation sets in earlier [S09]. This is why a two-parameter non-linearity model (light yield and effective pixel number) is used in the sPHENIX simulation of the EMCal response [S15].

Worked example: size of the saturation effect (toy)

One tower has $4 \times 40\,000 = 160\,000$ pixels. With a toy light yield of 400 p.e./GeV and a 20 GeV photon depositing 80% of its energy in the central tower, $N_{\mathrm{pe}} = 6400$. The linear approximation gives a deficit of $N_{\mathrm{pe}}/(2N_{\mathrm{pix}}) = 2\%$. If only a quarter of the pixels is effectively illuminated, the deficit becomes about 8%. Saturation is therefore a percent-level effect at the highest photon energies in sPHENIX and must be in the simulation. (Toy numbers.)

5.3 Temperature dependence

V_{bd} increases with temperature. At fixed bias, a higher temperature means a smaller overvoltage $V_{\mathrm{bias}} - V_{\mathrm{bd}}$ and therefore a smaller gain and PDE. The effect is a few percent per degree Celsius for typical operating overvoltages [S09, S16]. sPHENIX handles it in two ways: the calorimeter electronics are cooled and temperature-monitored, and the HCal calibration includes a run-by-run temperature correction (Section 10) [S16].

5.4 Dark counts and radiation damage

Thermal electrons can trigger avalanches without light: these are dark counts, at a rate of order hundreds of kHz per mm^2 at room temperature. Neutrons damage the silicon lattice: an incoming particle knocks a silicon atom out of its site (displacement damage, quantified by the non-ionising energy loss, NIEL), creating defects that generate dark current [S09]. Dark current and dark-count rate grow with neutron fluence, and the recovery (annealing) is slow and small [S09]. For a calorimeter this means a slowly rising pedestal noise, i.e. a growing noise term c/E in Eq. (7). sPHENIX was the first large deployment of SiPMs exposed to a significant neutron fluence in a collider experiment [S09].

Connect to your analysis

Rising SiPM noise over the years raises the pedestal width and therefore the energy below which towers are zero-suppressed or treated as noise. In your analysis this enters through the minimum cluster energy and the cluster χ^2/probability: if you compare Run-24 with later runs, check that the noise level has not shifted your low-p_T photon efficiency.

6 Hadronic showers

Goal. Understand why hadronic calorimetry is harder than electromagnetic calorimetry: larger and more fluctuating showers, invisible energy, $e/h \neq 1$, leakage.

6.1 Nuclear interaction length

A hadron travels on average one **nuclear interaction length** λ_I before an inelastic nuclear collision. λ_I scales roughly as $A^{1/3}$ and is much longer than X_0 for heavy materials: for iron $\lambda_I \approx 16.8$ cm while $X_0 \approx 1.76$ cm; for tungsten $\lambda_I \approx 9.9$ cm while $X_0 \approx 0.35$ cm [R4]. A hadronic calorimeter must therefore be many times deeper than an electromagnetic one. The sPHENIX EMCal, at about 18 X_0, is less than one λ_I deep (about 0.7 λ_I) [S08]. About half of the hadrons start their shower in the EMCal and the rest in the HCal.

6.2 Composition of a hadronic shower

In each nuclear interaction the hadron produces secondary pions, nucleons and nuclear fragments.

- **Electromagnetic component.** Neutral pions decay immediately to two photons, which shower electromagnetically. Once energy has gone into the EM component it never comes back (a π^0 does not make hadrons), so the EM fraction $f_{\rm em}$ grows with each generation. A common parametrisation is $f_{\rm em} = 1 - (E/E_0)^{k-1}$ with $E_0 \approx 1$ GeV and $k \approx 0.82$ for pions [R6]. This gives $f_{\rm em} \approx 0.25$ at 5 GeV and ≈ 0.46 at 30 GeV.
- **Hadronic component.** Charged hadrons ionise; neutrons travel far and deposit energy late; a sizeable fraction of the energy goes into breaking nuclei (binding energy) and is *invisible*.

6.3 Non-compensation, e/h and π/e

Let e be the response to the EM component and h the response to the non-EM component, per unit energy. Because of invisible energy, typically $h < e$. The response to a pion relative to an electron of the same energy is

$$\frac{\pi}{e} = f_{\rm em} + (1 - f_{\rm em})\frac{h}{e}. \tag{11}$$

Since $f_{\rm em}$ rises with energy, π/e rises with energy: a non-compensating calorimeter ($e/h \neq 1$) is non-linear for hadrons. Since $f_{\rm em}$ fluctuates event by event, the response fluctuates: this adds a large stochastic term and a constant term to the hadron resolution. With a toy $h/e = 0.6$, π/e goes from about 0.70 at 5 GeV to about 0.78 at 30 GeV. Good hadronic resolution requires $e/h \to 1$ ("compensation") [S09]. sPHENIX does not attempt compensation; the jet energy scale is corrected in software (Module 7).

6.4 Shower size and containment

Hadronic showers are long and wide. Empirical rules [R6]: the depth of maximum is $t_{\rm max} \approx 0.2 \ln E[\mathrm{GeV}] + 0.7$ (in λ_I), and 95% longitudinal containment needs $L_{95} \approx t_{\rm max} + 2.5\,\lambda_{\rm att}$ with $\lambda_{\rm att} \approx \lambda_I (E/\mathrm{GeV})^{0.13}$.

Worked example: how deep must the sPHENIX calorimeter be?

For a 20 GeV hadron: $t_{\rm max} = 0.2 \times 3.0 + 0.7 = 1.3\,\lambda_I$; $\lambda_{\rm att} = 20^{0.13} = 1.48\,\lambda_I$; $L_{95} \approx 1.3 + 3.7 = 5.0\,\lambda_I$. For 10 GeV: $4.5\,\lambda_I$; for 50 GeV: $5.6\,\lambda_I$. The full sPHENIX stack (EMCal $\approx 0.7\lambda_I$, inner HCal, solenoid, outer HCal $\approx 3.5\lambda_I$) is roughly $5\lambda_I$ [S08, R3]. So it contains about 95% of a 20 GeV hadron shower on average, and a 50 GeV hadron leaks noticeably. The design target was 95% containment around 20–30 GeV, and the calorimeters are "a little thin" by design, so leakage at high energy is real and must come from simulation [S08].

Pitfalls

- Calling the HCal "calibrated" without saying to *what*: sPHENIX calibrates the HCal to the *electromagnetic scale* (Section 10). A hadron then reads low by $\pi/e < 1$, and the jet energy scale correction restores it.
- The three longitudinal compartments are made of different materials (tungsten EMCal, aluminium inner HCal, steel outer HCal), each with its own e/h and sampling fraction [S08]. A hadron's measured energy depends on where it starts showering.

7 The sPHENIX HCal

Goal. Know the geometry and construction of the inner and outer HCal and the reasons for the tilted-plate design.

7.1 Overview

The HCal is the first mid-rapidity sampling hadronic calorimeter at RHIC [S16]. It has two layers:

- **Inner HCal (iHCal):** inside the solenoid, directly behind the EMCal, at a radius of roughly 1.2–1.4 m. Its absorber is mostly aluminium; it also serves as the support frame for the EMCal [S08].
- **Outer HCal (oHCal):** outside the solenoid, at a radius of roughly 1.9–2.6 m. Its absorber is tapered low-carbon steel plates which are also the magnet flux return [S08].

Each layer has 24×64 towers in $\eta \times \phi$, i.e. 1536 towers, with $\Delta\eta \times \Delta\phi \approx 0.1 \times 0.1$, covering $|\eta| < 1.1$ and 2π [S08, S16]. Each layer is built from 32 sectors in ϕ; one sector is 2 towers wide in ϕ and 24 towers long in η [S16].

7.2 Tiles, fibres and SiPMs

The active medium consists of extruded plastic scintillator tiles with an embedded wavelength-shifting (WLS) fibre in a groove [S08, S16]. The WLS fibre absorbs the blue scintillation light and re-emits it at a longer wavelength, which is trapped and guided to one end of the fibre, where a 3×3 mm^2 SiPM reads it out [S08]. Each tower combines the signals of several tiles in depth: 5 tiles per tower in the oHCal and 4 in the iHCal [S16]. In total there are $1536 \times 5 = 7680$ tiles/SiPMs in the oHCal and $1536 \times 4 = 6144$ in the iHCal [S16]. The tile shapes vary with η (projective geometry), so there are many different tile types; tile production and quality control (light yield, uniformity) were a large part of the project [S08]. The response across each tile was *mapped* with beam at Fermilab, and rules of thumb for the fibre routing were developed to make the light collection as uniform as possible [S08].

7.3 Why tilted plates?

In a plain radial geometry, a particle from the vertex could travel along a gap between plates and never sample the active medium ("channelling"). Tilting the plates with respect to the radial direction means that every straight track from the vertex crosses several plates and tiles [S08]. The tilt also lets the plates double as an efficient magnetic flux return [S08]. Two side effects had to be studied: a small asymmetry between positively and negatively charged particles (which bend in opposite directions in ϕ and therefore cross the tilted plates at different angles), and a sampling fraction that changes with depth [S08].

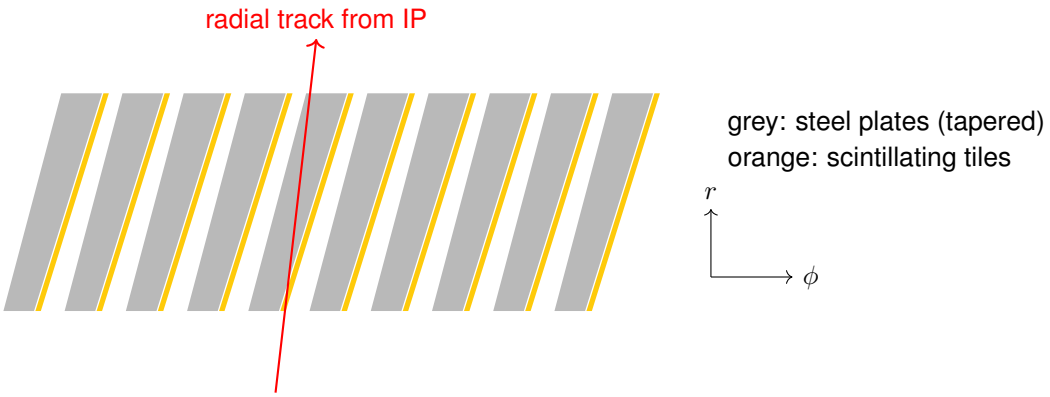


Figure 5: Schematic of the tilted-plate HCal in the r–ϕ plane: a radial track always crosses several absorber plates and tiles. Not to scale; tilt angle exaggerated.

7.4 Depth budget

Approximate depths along a radial line at $\eta = 0$ [S08]:

Element	Material	Depth
EMCal	W powder/epoxy + fibres	$\approx 18X_0 \approx 0.7\lambda_I$
Inner HCal	mostly aluminium + tiles	small fraction of λ_I
Solenoid	superconducting coil, cryostat	$\approx 1.4X_0$
Outer HCal	steel + tiles	$\approx 3.5\lambda_I$

The original inner HCal design used steel; the as-built inner HCal is an aluminium frame [S08, S47]. This matters for comparing with the 2016–2018 beam tests, which had a steel inner HCal (Section 9).

7.5 The LED system

Each HCal channel can be flashed with an LED. The original hope was to track gains with LEDs alone [S08]. In practice the coupling of the LED to each tile is variable, the light is injected at a single point close to the SiPM (so it does not sample the tile response), the LED pulser stability itself must be characterised, and in the oHCal the LED signals of the five tiles of a tower are summed [S08]. LEDs are therefore excellent for channel checks, timing and *relative* gain changes (e.g. the temperature coefficient), but not for the absolute energy scale. The absolute scale comes from physics signals: cosmic muons and collision data (Section 10).

Checks

$1536 = 24 \times 64$ towers per layer; 0.1×0.1; 32 sectors per layer; 5 tiles/tower in oHCal, 4 in iHCal. One HCal tower $\leftrightarrow 4 \times 4$ EMCal towers. In a tower-occupancy map, look for the sector structure every 2 towers in $i\phi$ and for special regions near support structures and the “chimney” (where the magnet services exit) [S16].

Connect to your analysis

Your analysis does not use the HCal energy directly, but it touches it in two ways. (1) Photon isolation and some photon-ID variables use the HCal energy behind the EMCal cluster: hadrons and merged π^0s can leave energy in the iHCal. (2) The jet triggers combine EMCal and HCal energy; if you look at Jet-trigger samples for cross-checks, the HCal calibration enters through the trigger threshold.

8 Readout chain: from light to a calibrated tower

Goal. Follow one tower signal from the SiPM to the `TowerInfo` object in your DST, and know which quantities you can trust.

8.1 Front end and digitisation

1. The SiPM signal is amplified and shaped by a preamplifier close to the detector. The EMCal readout boards are mounted with the SiPMs on the sector; the HCal uses similar SiPM preamplifier boards [S09, S08].
2. The analog signals travel to digitiser crates in racks on the detector platform. Large SiPM signals made remote digitisation possible [S08].

3. Each channel is sampled by a 14-bit flash ADC at six times the RHIC beam clock, about 56 MHz [S08, S15]. The RHIC bunch spacing is about 106 ns ($1/(78.2\ \mathrm{kHz} \times 120)$ buckets), so one sample is about 17.7 ns, and a pulse spans several samples.
4. On a trigger (Module 5), a short window of samples is read out per channel (16 samples in the 2024–2026 runs [S15]).

8.2 Zero suppression

Most towers in a p+p event see no signal. To save bandwidth, a channel is *zero suppressed*: only two samples are kept (an early, pre-pulse sample and the sample at the expected peak), unless the difference between them exceeds a threshold, in which case the full waveform is kept [S15]. For a zero-suppressed channel the energy estimate is just the difference of the two samples, and time and χ^2 are undefined [S15].

8.3 Waveform fitting

For full waveforms, the offline module `CaloTowerBuilder` (coresoftware `offline/packages/CaloReco`) fits a template pulse shape to the samples [S15]. The fit returns:

- the **amplitude** (peak height above pedestal, in ADC counts);
- the **peak time** (in units of samples);
- the **pedestal**;
- a **fit-quality** χ^2 per degree of freedom;
- an 8-bit **status** word.

These "distilled" tower quantities are written to the production DSTs as a `TowerInfo` container (node `TOWERS_CEMC` for the EMCal; similar nodes for the HCal and sEPD) [S15]. The tower values in the DST are *uncalibrated*, in ADC counts [S15]. Several `TowerInfo` versions exist (v1 saves only the energy; later versions add time, χ^2, pedestal and, optionally, the waveform); the DSTs use a compact version with reduced precision for time and χ^2 [S15]. Check the current coresoftware source for the exact layout.

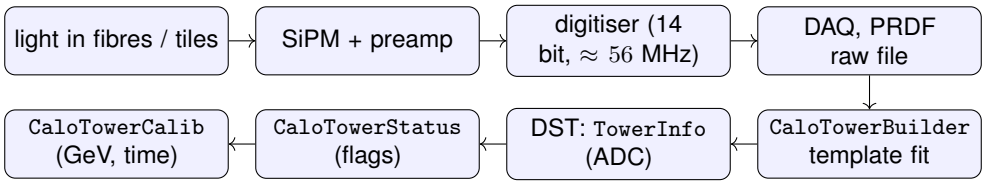


Figure 6: Schematic calorimeter signal chain. The top row is hardware (Modules 3 and 5); the bottom row is software (Modules 6 and 7). Status and calibration are applied when you run the calorimeter reconstruction macro on the DST [S15].

8.4 Status flags and calibration on the fly

When you run the standard calorimeter macro (`Process_Calo` in the macros repository) on a DST, two modules run [S15]:

- `CaloTowerStatus` sets per-tower flags, for example: hot/dead/cold (from an offline comparison of towers at the same η, with 5σ outliers flagged), bad χ^2, not instrumented, no calibration available, zero suppressed, saturated (the waveform reached the ADC maximum) [S15].

- `CaloTowerCalib` multiplies each tower amplitude by a single calibration constant (separate constants for zero-suppressed and full-waveform amplitudes) to convert ADC to GeV, and subtracts the run-averaged tower time so that calibrated times are centred near zero [S15].

The constants come from the conditions database (CDB) through `CDBInterface`, selected by a global tag and the run number (Module 6). In the EMCal, "GeV" is defined such that a single photon has a response of 1, i.e. it is a photon energy scale [S15].

Pitfalls

- Using the DST `TOWERS_CEMC` values directly as energies: they are ADC counts. Use the calibrated node produced by the macro (in your analysis, `TOWERINFO_CALIB_CEMC`).
- Ignoring the status word: hot towers make fake high-p_T clusters; bad-χ^2 towers have distorted pulses (e.g. pile-up or electronics problems) [S15].
- Mixing zero-suppressed and full-waveform towers without care: their energy estimators differ (two-sample difference vs. template amplitude), and they have separate calibration constants [S15].
- Timing is reported in samples, not ns, and has an energy dependence that is not fully understood [S15].

9 Test beams and simulation

Goal. Know what the test beams established, how to read their results critically, and why simulation is the bridge from test beam to sPHENIX.

9.1 The test-beam campaigns

Prototypes were tested at the Fermilab Test Beam Facility as experiment T-1044 [S09, S08, S47, R1, R2]:

- 2014: proof-of-principle prototypes.
- 2016: prototypes at $\eta \approx 0$ (EMCal + inner HCal + outer HCal in all combinations; the inner HCal was steel). Published in [R1].
- 2017: prototypes at $\eta \approx 0.9$ (high-η geometry).
- 2018: improved high-η EMCal blocks (a 2D-projective prototype built like a real sector in mechanics, electronics and cooling) [S09]; EMCal results published in [R2]. Some data were also taken with an uninstrumented aluminium inner-HCal frame [S47].

Beam energies ranged up to about 30 GeV, with mixed electron/hadron beams; Cherenkov counters tagged electrons, and wire chambers or hodoscopes measured the beam position [R1].

9.2 Combining compartments

In a test beam with several longitudinal compartments, the hadron energy is reconstructed as a weighted sum, for example $E = c\,[g\,E_{\mathrm{EMCal}} + w\,E_{\mathrm{iHCal}} + E_{\mathrm{oHCal}}]$. The weights correct for different sampling fractions and e/h in each compartment. In the published 2016 analysis the inner-HCal signal was given a larger weight than the outer-HCal signal in data, and the EMCal weight was tuned versus the EMCal energy fraction to flatten the response [S47, R1].

9.3 Lessons from re-analysing the test beam

A 2026 workfest study re-ran the original prototype Geant4 simulation (code in the public sPHENIX `prototype` repository) to compare data and simulation in a way that could constrain the single-hadron energy scale used in jet calibrations [S47]. The general lessons, which apply to any test-beam or calibration plot you read:

1. Check the normalisation. If each energy point is scaled to the beam energy separately, only widths are compared, not linearity or absolute scale.
2. Check that the geometry in the simulation matches the hardware that was in the beam (there were two different steel inner-HCal prototypes, and a mismatch was found and fixed by comparing Geant4 hit files) [S47].
3. Check the beam composition at low energy: muon or other contamination can bias the low-energy points [S47].
4. Weights must be the same in data and simulation for an apples-to-apples comparison [S47].
5. The final sPHENIX inner HCal is aluminium, not steel. Leakage differs from the prototype, so the full sPHENIX geometry must be simulated and the leakage modelling varied as a systematic [S47].

9.4 Simulation in sPHENIX

The full detector is simulated with Geant4 inside Fun4All through `PHG4Reco` (Module 6). The calorimeter geometries are set up in the standard macros (in the `macros` repository, `common/` directory, e.g. `G4_CEmc_Spacal.C`, `G4_HcalIn_ref.C`, `G4_HcalOut_ref.C`; check the current names). Geant4 deposits are converted to light (with the sampling structure and, for the EMCal, a light-collection map), then to waveforms that mimic the real electronics, so that the same `CaloTowerBuilder` reconstruction runs on data and simulation. EMCal SiPM non-linearity was included in the test-beam simulation and a two-parameter model has been implemented for sPHENIX simulation; samples must be re-digitised to include it [S15].

Connect to your analysis

Your efficiency uses PYTHIA8 events through the sPHENIX simulation plus extra EMCal smearing tuned on data. Three hardware effects in this module must be correctly in that simulation or covered by systematics: (1) lateral and longitudinal leakage (energy scale per cluster); (2) SiPM non-linearity (high-p_T photons); (3) the dead/hot tower map (acceptance). For (3) the simulation should use the same tower-status map as the data, or you should apply the data map to the simulated towers.

10 Calibration strategy overview

Goal. Understand the logic of the calibration chain for both calorimeters: what each step fixes, what physics it relies on, and what it cannot fix. The details of the code are in Module 7.

10.1 What a calibration must do

For each tower i we need a constant k_i such that $E_i = k_i\, A_i$, where A_i is the amplitude in ADC counts. It is useful to split $k_i = K \cdot r_i(t)$:

- an **absolute scale** K (common to all towers, or to all towers at a given η), which makes the energy correct on average;
- a **relative** (inter-tower) factor r_i, which equalises towers;

- a **time dependence** $r_i(t)$, from temperature, radiation damage and hardware changes.

Errors in r_i broaden peaks (they enter the constant term b); errors in K shift peaks and spectra.

10.2 Tools available

Tool	What it measures	Limitations
Test beam	absolute response, resolution, linearity for prototypes	different hardware and environment
LED	channel alive, timing, relative gain changes	no absolute scale; coupling varies [S08]
Cosmic muons (MIP)	energy deposit of a known particle in each tower	must be converted to the shower scale with simulation; muons are mostly vertical [S16]
Tower slope	relative gain from the shape of each tower's energy spectrum in beam data	needs symmetry (all towers at one η see the same spectrum) [S42, S16]
$\pi^0 \to \gamma\gamma$	absolute photon scale and tower-by-tower gains (EMCal)	needs statistics, depends on resolution and non-linearity [S15]
Isolated hadrons, E/p	hadron response with tracks	needs tracking; planned [S16]

10.3 The tower-slope method

Idea. In minimum-bias data, all towers at the same η should see the same energy spectrum, because the physics is symmetric in ϕ. If tower i has a gain g_i relative to the average, its spectrum is a stretched copy of the reference.

Derivation. Let the true spectrum be $F(E)$. The measured spectrum of tower i is

$$f_i(E') = \frac{1}{g_i} F\left(\frac{E'}{g_i}\right), \qquad E' = g_i E. \tag{12}$$

For a locally exponential spectrum $F(E) \propto e^{-E/T}$, $f_i(E') \propto e^{-E'/(g_i T)}$: the gain simply multiplies the inverse slope. So the ratio of fitted slopes gives g_i directly. In practice the reference is the sum of all tower spectra in the same η ring, and a two-parameter function (normalisation and gain stretch) built from the reference is fitted to each tower's spectrum [S42, S16]. The low-energy region is excluded, because noise, pedestal fluctuations, zero suppression and trigger thresholds distort it [S42].

Why it is useful. It needs no special run and no reconstruction beyond towers. It is applied as a relative correction on top of an absolute calibration (cosmics in the HCal), so it should leave the overall scale unchanged; a check that the averaged correction preserves the absolute scale is part of the validation [S42].

Where it fails. Towers where the ϕ symmetry is broken by material: support structures and the chimney region. These are excluded from the method [S16].

Link to your M.S. work

Your Tower Slope calibration MC studies tested exactly this method for the EMCal: inject known gain offsets in simulation, run the slope fits, and check how well the injected g_i are recovered as a function of statistics and fit range. The two lessons from Section 10—exclude the low-energy region and respect the symmetry assumption—are the same ones you met there.

10.4 HCal: cosmics, tower slope, temperature, special cases

The HCal calibration has four steps [S16]:

1. **Absolute scale from cosmic muons.** Dedicated cosmic runs select muons that pass vertically through a whole tower (topological cuts using neighbouring towers). The deposited-energy spectrum of each tower is fitted (a gamma-type function) to find its most probable value (MPV) in ADC. The same is done in a cosmic-muon simulation, giving the MPV in MeV. The ratio converts ADC to energy at the *electromagnetic* scale, tower by tower.
2. **Relative correction by tower slope** with beam data (above), for effects not in the cosmic simulation (e.g. structures outside the main detector body).
3. **Run-by-run temperature correction**, using the SiPM gain temperature coefficient (measured with LEDs) and the recorded temperature, to transport the cosmic calibration to each physics run.
4. **Special corrections** for rare, correctable channel pathologies, e.g. a tower whose signal is halved or doubled for a period because of a cable connection problem; these are found by inspecting per-tower hit rates versus run number.

Validation. The calibration stability is checked with the hit rate per tower per event versus time. Because this rate grows with the probability of a second collision in the same crossing, it must be corrected for pile-up [S16].

Worked example: pile-up probability per crossing

RHIC revolution frequency ≈ 78 kHz; with 111 filled bunches per ring, colliding crossings occur at $f_{\rm bc} \approx 8.7$ MHz [S16]. For an interaction rate R, the mean number of collisions per crossing is $\mu = R/f_{\rm bc}$, and for Poisson statistics the probability of two or more collisions given at least one is $P(\geq 2| \geq 1) = [1 - e^{-\mu}(1+\mu)]/(1-e^{-\mu}) \approx \mu/2$. Toy rates: $R = 10$ kHz gives $\mu = 0.0012$ and $P \approx 0.06\%$; $R = 300$ kHz gives $\mu = 0.035$ and $P \approx 1.7\%$. The rate-dependent part of the hit rate must be removed before using it as a gain monitor. In practice a data-driven correction may work better than the simple Poisson formula [S16]. (Toy rates.)

10.5 EMCal: π^0 gain balancing and absolute scale

The EMCal calibration uses the $\pi^0 \to \gamma\gamma$ peak [S15]:

1. **Start** from an initial set of constants (from lab tests, cosmics, or a previous calibration).
2. **Gain balancing.** For each tower, fill the diphoton mass spectrum for pairs where one cluster's leading tower is that tower. Fit the π^0 peak. Correct the tower gain by the ratio of the target to the fitted mass. Iterate, because changing one tower's gain changes clusters that share it with neighbours.
3. **Absolute scale.** Match the peak position to simulation, not to the PDG mass, because the reconstructed peak is shifted by resolution, leakage, the clustering and the kinematic selection. The absolute scale is sensitive to the single-photon resolution, so the MC cluster energies and positions are smeared until the data and MC π^0 and η peaks agree [S15].
4. **Timing calibration.** Subtract each tower's mean time so that prompt photons sit at zero; the photon timing resolution can be checked with the time difference of the two π^0 decay photons [S15].
5. **Non-linearity.** Model the SiPM and electronics non-linearity in simulation (two-parameter model) [S15].

The tower-slope method can also be used for the EMCal as a fast relative calibration or a starting point.

Worked example: why the peak is not at 135 MeV

Suppose each photon's measured energy is $E' = (1-\ell)E$, with leakage $\ell = 5\%$ not absorbed in the calibration. Then $m' = \sqrt{2E_1'E_2'(1-\cos\theta)} = (1-\ell)m$, and the peak moves to about 128 MeV. Now add resolution: a falling p_T spectrum feeds more low-energy pairs upward into a fixed p_T bin than the reverse, so pairs that fluctuated up in energy are over-represented, and the peak in a given p_T bin moves *up*. The two effects partly cancel, by an amount that depends on p_T. Hence the calibration target is "peak position in data equals peak position in MC", not "peak equals the PDG mass". (Toy reasoning.)

Checks

- After calibration, the tower-by-tower π^0 peak positions should scatter much less than before; their spread is a direct estimate of the remaining relative miscalibration.
- A tower-occupancy map at fixed η should be flat in ϕ except for known structures.
- The hit rate per event versus run (pile-up corrected) should be flat after the temperature correction.
- The calibrated time of prompt clusters should peak at zero.

Pitfalls

- Calibrating relative gains with the tower-slope method in a region where ϕ symmetry is broken (supports, chimney, sector edges with dead neighbours).
- Using a trigger-selected sample for the tower slope: the trigger threshold biases the spectrum shape tower by tower. Use minimum-bias data or fit only far above threshold.
- Calibrating the π^0 peak to the PDG mass in data and also in MC with different procedures: the data/MC energy scale is then inconsistent.
- Forgetting the time dependence: a calibration from one period applied to another period without temperature/run corrections.

Connect to your analysis

- The global tag and the calibration version used for `DST_CALOFITTING`-based reconstruction must be the same for all runs in your cross section, or you must show the difference is negligible.
- The residual tower-to-tower miscalibration enters your π^0 width (constant term). It is one of the reasons the MC smearing is needed.
- The Photon triggers (bits 29 and 30 in your run period) fire on EMCal energy sums computed from *online* tower energies with *online* gains; their turn-on curves in offline energy are therefore smeared by the offline/online calibration difference (Module 5).

11 Summary

- EM showers scale with X_0 (depth) and R_M (width); the sPHENIX EMCal has $X_0 \approx 7$ mm, $R_M \approx 2.3$ cm and $\approx 18X_0$ depth, and a 2 GeV photon is about 91% contained in a 3×3 array (toy estimate).
- The EMCal is a W-powder/epoxy SPACAL with 0.47 mm fibres, $f_{\rm samp} \approx 2\%$, 96×256 projective towers read out by four SiPMs each; the sampling term alone gives about $13\%/\sqrt{E}$.

- The HCal has two layers of 24×64 towers (0.1×0.1): aluminium inner HCal inside the solenoid and steel outer HCal (the flux return) outside, both with tilted plates and tile+WLS+SiPM readout.
- Public beam-test resolutions: EMCal $2.8\% \oplus 15.5\%/\sqrt{E}$ at $\eta \approx 0$; hadrons in EMCal+HCal $13.5\% \oplus 64.9\%/\sqrt{E}$.
- Signals are digitised at six times the beam clock, fitted offline by `CaloTowerBuilder`, written as uncalibrated `TowerInfo`, and flagged/calibrated on the fly by `CaloTowerStatus` and `CaloTowerCalib`.
- Calibration: cosmics (HCal absolute), tower slope (relative), temperature (time), π^0 (EMCal relative and absolute), with MC smearing tuned to data.

12 Hands-on exercises on SDCC (optional, suggested)

These exercises are optional. They have not been run as part of writing this module; commands and names are suggestions that you should check against the current software. The text above does not depend on them.

1. **Draw the EMCal tower grid.** On paper or in ROOT, draw a 96×256 `TH2` in $(i\eta, i\phi)$; overlay block boundaries every 2 towers, sector boundaries every 8 towers in $i\phi$ and at $i\eta = 48$, and the 24×64 HCal grid. Convert tower indices to η and ϕ by $\eta \approx -1.1 + (i\eta + 0.5) \times 2.2/96$ and $\phi = (i\phi + 0.5) \times 2\pi/256$ (approximate; the real geometry is in the geometry node).
2. **Containment calculator.** Write a 20-line Python or ROOT script that evaluates $P(a, bT)$ for the gamma profile and reproduces the containment numbers of Section 2 for $E = 0.5$–30 GeV. Then vary E_c from 8 to 12 MeV to see how sensitive the estimate is.
3. **Tower map from a DST.** Set up the environment, take one `DST_CALOFITTING` file of a Run-24 p+p run from the FileCatalog (Module 5), run `Process_Calo()`, and in a small `SubsysReco` loop over `TOWERINFO_CALIB_CEMC` fill an occupancy map for towers above 0.5 GeV and a map of towers flagged hot. Identify the sector boundaries. Environment set-up (suggested):

   ```
   source /opt/sphenix/core/bin/sphenix_setup.sh -n new
   ```

4. **Look up one tower's calibration in the CDB (checkpoint).** In a ROOT macro, set the global tag and run number through `recoConsts` (flags `CDB_GLOBALTAG` and `TIMESTAMP`), get the payload URL from `CDBInterface`, open it with `CDBTTree`, call `LoadCalibrations()`, and read the value for one tower key (encode the key with `TowerInfoDefs`). A sketch (check the current calibration names and key encoding in coresoftware before running):

   ```
   std::string url = CDBInterface::instance()->getUrl("CEMC_calib_ADC_to_ETower");
   CDBTTree *t = new CDBTTree(url);
   t->LoadCalibrations();
   unsigned int key = TowerInfoDefs::encode_emcal(ieta, iphi);
   float k = t->GetFloatValue(key, "CEMC_calib_ADC_to_ETower");
   ```

5. **Tower-slope toy.** Generate exponential spectra for 256 toy towers with random gains of a few percent, fit each against the summed reference above a threshold, and plot recovered vs. injected gain. Repeat with a threshold bias to see the pitfall.

13 Self-test questions

1. Why do photons reach shower maximum about one radiation length later than electrons?
2. Estimate $t_{\max}$ for a 10 GeV photon in the sPHENIX EMCal, and the fraction of energy leaking out of an 18 X_0 deep block.

3. Why must the EMCal be dense and finely segmented for Au+Au physics? Give a quantitative argument.
4. What is the sampling fraction of the EMCal, and why is it smaller for EM showers than the MIP value?
5. Using the rule of thumb, estimate the stochastic term of a sampling calorimeter with 1 mm fibres and 5% sampling fraction. Is it better or worse than sPHENIX?
6. For $\sigma_E/E = 2.8\% \oplus 15.5\%/\sqrt{E}$, at which energy are the stochastic and constant terms equal?
7. Explain SiPM saturation and how it affects a high-energy photon. Why does poor light mixing make it worse?
8. What happens to a SiPM's gain when the temperature rises at fixed bias, and how does sPHENIX correct for it in the HCal?
9. Why is a non-compensating calorimeter non-linear for hadrons?
10. Why are the HCal plates tilted? Name one side effect that had to be studied.
11. A DST tower in `TOWERS_CEMC` has amplitude 250. What energy is that, and what do you need to know to convert it?
12. What is a zero-suppressed tower, and what are its energy, time and χ^2?
13. State the assumption behind the tower-slope method and two places in the HCal where it fails.
14. Why is the EMCal absolute scale tuned by matching the π^0 peak to MC rather than to 135 MeV?
15. Why can LEDs not provide the absolute HCal scale?

A Answers

1. A photon must first convert into an e^+e^- pair, with mean free path $\frac{9}{7}X_0$; until then nothing is deposited. Electrons start radiating immediately. The parametrisation encodes this as $C_\gamma - C_e = +1$.
2. $t_{\max} = \ln(10000/8) + 0.5 = 7.1 + 0.5 \approx 7.6\,X_0$ (about 5.3 cm). With $b = 0.5$, $a = 1 + 0.5 \times 7.6 = 4.8$, and $P(4.8, 9) \approx 0.95$, so about 5% leaks out of the back (pure-tungsten E_c; somewhat less for the real mixture).
3. The underlying event deposits energy in every tower. Its contribution to a cluster scales with the cluster area. With $\mathrm{d}E_T/\mathrm{d}\eta \sim 600$ GeV, a 3×3 array (≈ 0.0055 in $\eta\phi$) collects ~ 0.5 GeV, comparable to the 0.34 GeV resolution for a 5 GeV electron. A larger R_M or larger towers would make the UE dominate. Density (small X_0 and R_M) also lets the calorimeter fit into 26 cm of radius.
4. About 2%. Soft photons in the shower are absorbed preferentially by the high-Z tungsten (photo-electric cross section $\propto Z^{4-5}$), so the active fibres see less than their MIP share.
5. $2.7\%\sqrt{1/0.05} = 2.7\% \times 4.47 = 12\%$. Slightly better than the sPHENIX $\approx 13\%$, but such a calorimeter would be less dense (more active material) and so need more space.
6. $15.5\%/\sqrt{E} = 2.8\%$ gives $\sqrt{E} = 5.5$, $E \approx 31$ GeV.
7. Each SiPM pixel fires at most once per pulse, so the number of fired pixels is $N_{\mathrm{pix}}(1 - e^{-N_{\mathrm{pe}}/N_{\mathrm{pix}}})$ and falls below N_{pe} as the light grows. A high-energy photon concentrates light in one tower, so its energy is underestimated. With poor mixing only part of the pixel area is illuminated, the effective N_{pix} is smaller, and saturation starts at lower energy.

8. The breakdown voltage rises, so the overvoltage, gain and PDE fall (a few percent per degree). The HCal calibration applies a run-by-run correction using the measured temperature and the LED-measured temperature coefficient, relative to the conditions of the cosmic calibration.
9. The EM fraction f_{em} of a hadron shower grows with energy, and the EM and hadronic components have different responses ($e/h \neq 1$). So $\pi/e = f_{\mathrm{em}} + (1 - f_{\mathrm{em}})h/e$ grows with energy: the response per GeV changes with energy.
10. So that every straight track from the vertex crosses several absorber plates and tiles (no channelling along gaps), and so that the steel can act as the flux return. Side effects: a charge-dependent ($\pm$) response because positive and negative tracks bend differently relative to the tilt; and a depth-dependent sampling fraction.
11. Nothing yet: it is 250 ADC counts above pedestal. You need the tower's calibration constant for the run (from the CDB via the global tag and run number), applied by `CaloTowerCalib`, plus its status (hot, bad χ^2, etc.). You also need to know whether it was zero suppressed, since that uses a separate constant.
12. A channel whose signal was below threshold, so only two samples were kept. Its energy is the difference of the two samples; time and χ^2 are not defined (NaN).
13. All towers at the same η see the same energy spectrum (ϕ symmetry). It fails near support structures and in the chimney region, where extra material changes the spectrum.
14. The reconstructed peak position is shifted by leakage, clustering, resolution and the kinematic selection in a falling spectrum. Matching data to MC, with MC smeared to the data resolution, makes the data and MC energy scales consistent, which is what an efficiency correction needs.
15. The LED light is injected at one point near the SiPM and does not sample the tile response; its coupling varies from channel to channel; and the LED pulse amplitude itself is not an absolute standard. LEDs measure relative changes well, not energy.

B Glossary

Radiation length X_0 Distance over which an electron loses $1 - 1/e$ of its energy by bremsstrahlung; $\frac{7}{9}$ of the photon pair-production mean free path.

Critical energy E_c Energy at which ionisation and radiation losses of electrons are equal; ends shower multiplication.

Molière radius R_M Transverse shower scale; $\approx 90\%$ of the energy lies within $1R_M$.

Nuclear interaction length λ_I Mean free path for an inelastic nuclear interaction.

Sampling fraction Fraction of deposited energy that lands in the active medium.

Stochastic/constant/noise term The $a/\sqrt{E}$, b and c/E terms of the resolution.

SPACAL Spaghetti calorimeter: scintillating fibres in an absorber.

Projective tower Tower whose axis points to the interaction point.

SiPM / MPPC Silicon photomultiplier, an array of Geiger-mode avalanche photodiodes.

PDE Photon detection efficiency.

Breakdown voltage Bias above which a pixel avalanches; rises with temperature.

WLS fibre Wavelength-shifting fibre that collects tile light and guides it to the SiPM.

e/h**,** π/e Ratio of electromagnetic to hadronic response; pion-to-electron response ratio.

Zero suppression Keeping only two samples for a channel below threshold.

Template fit Fit of a reference pulse shape to the waveform samples.

`TowerInfo` The tower object (energy, time, χ^2, status) in sPHENIX DSTs.

Tower slope Relative calibration from the shape of each tower's spectrum, assuming ϕ symmetry.

MIP / **MPV** Minimum-ionising particle / most probable value of its deposit.

CDB, global tag Conditions database and the label that selects a consistent set of calibrations.

Chimney Region of the detector where the magnet services exit; non-standard material.

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